Lesson notes · for you

Fair turns without clashes

Optimization · Level 1 · Lesson 4

Time	Materials
20 min	Paper; Three or four coloured pencils

Learning goals

- Draw clashes as dots and lines.
- Colour the dots so nothing clashes, with as few colours as you can.
- Share turns fairly.

Activity: Colour the clashes

1. Draw (5 min): Draw eight dots in a circle and label them: maths, English, science, Arabic, history, geography, art, French.
2. Join (4 min): Join the tests each pupil takes: Sami (English, science), Hala (history, geography, art), Omar (maths, Arabic, French), Dina (Arabic, art), Nadia (science, French), Tarek (science, Arabic, French).
3. Colour (7 min): Colour each dot so that joined dots never share a colour. Start with the hardest dot. How few colours did you need?
4. Share (4 min): Compare with a partner. Did you both need three colours? Find Omar's three tests: why can't two colours be enough?

Use coloured pencils, or write the slot numbers 1, 2, 3 in the dots. No device needed.

Discussion

- Why do Omar's three tests need three times?
- Why does starting with the hardest dot help?
- Which turn at school would you share more fairly?

Check yourself

- A line between two dots means ___. The fewest times for our test week is ___.

Try it this week

Draw your class's clubs or games as dots, and join two dots when a friend does both. Colour the dots with as few colours as you can, so joined dots never share a colour. Then make a fair rota for one class job, so everyone takes a turn.

Worksheet

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Name _____ Class _____ Date _____

1 **Omar takes maths, Arabic and French, and each pair of them is joined by a line. Can the school fit the whole test week into two times?**

- ◯ Yes, with clever colouring
- ◯ No: his three tests need three different times
- ◯ Yes, if he takes them faster

2 **If two tests are joined by a line, they can be at the same time.**

- ◯ True
- ◯ False

3 **Match each part of the drawing to what it means.**

A dot	• • • •	A pupil takes both tests
A line	• • • •	At least three times needed
A colour	• • • •	A test
Three dots all joined	• • • •	A time slot

4 **Maths is joined by lines only to Arabic and French. Which tests can be at the same time as maths? Choose all that apply.**

- ☐ French
- ☐ History
- ☐ English
- ☐ Arabic
- ☐ Art

5 **Make a fair rota for cleaning the board. Put the steps in order.**

- ☐ Move one name down the list each day
- ☐ Write everyone's name on a list
- ☐ When the list ends, start again at the top
- ☐ Give the job to the first name on Monday

✎ **Think of a turn at school or at home that some people always seem to get: the worst seat, the hardest job, the last go. How could you share it more fairly?**

Check your answers

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1 **No: his three tests need three different times**

Maths, Arabic and French are all joined to each other, so no two of them can share a time.

2 **False**

A line means a pupil takes both tests, so they clash: they need different times (different colours).

3 **A dot — A test**
A line — A pupil takes both tests
A colour — A time slot
Three dots all joined — At least three times needed

Dots are tests, lines are clashes and colours are times.

4 **English; Art; History**

Arabic and French are joined to maths, so they clash with it.

5 **Top to bottom: 3, 1, 4, 2**

Going down the list and starting again means everyone has one turn before anyone has two: fair for everyone.